

AWS EC2 Monitoring with CloudWatch Alarms and SNS Notifications

This project demonstrates operational monitoring and alerting workflows in AWS using Amazon EC2, Amazon CloudWatch, and Amazon SNS. Simulated high CPU utilization through an EC2 terminal session to validate CloudWatch alarm thresholds and SNS notifications.

Architecture Workflow

- EC2 Instance
- CloudWatch Metrics
- CloudWatch Alarm
- Amazon SNS
- Email Notification

Figure 1 — Amazon EC2 instance configured for CPU monitoring.

An EC2 instance **CloudWatch Trigger Instance** was launched to simulate infrastructure monitoring and operational alerting workflows.

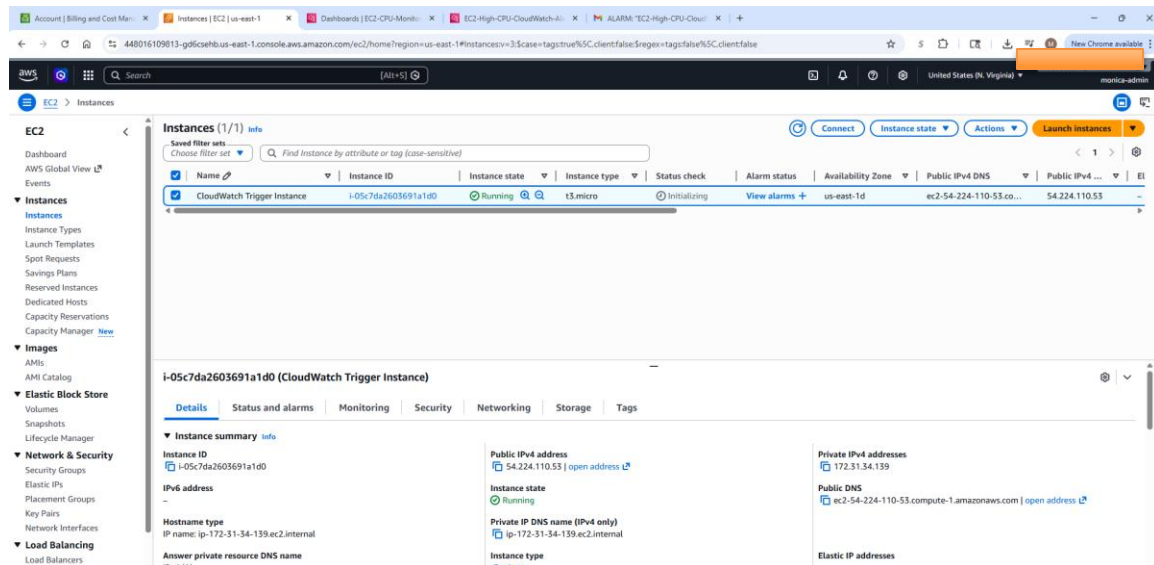


Figure 2 — CloudWatch dashboard monitoring EC2 CPU utilization metrics.
CloudWatch dashboard widgets displayed real-time CPU utilization metrics for the EC2 instance **before** spike.

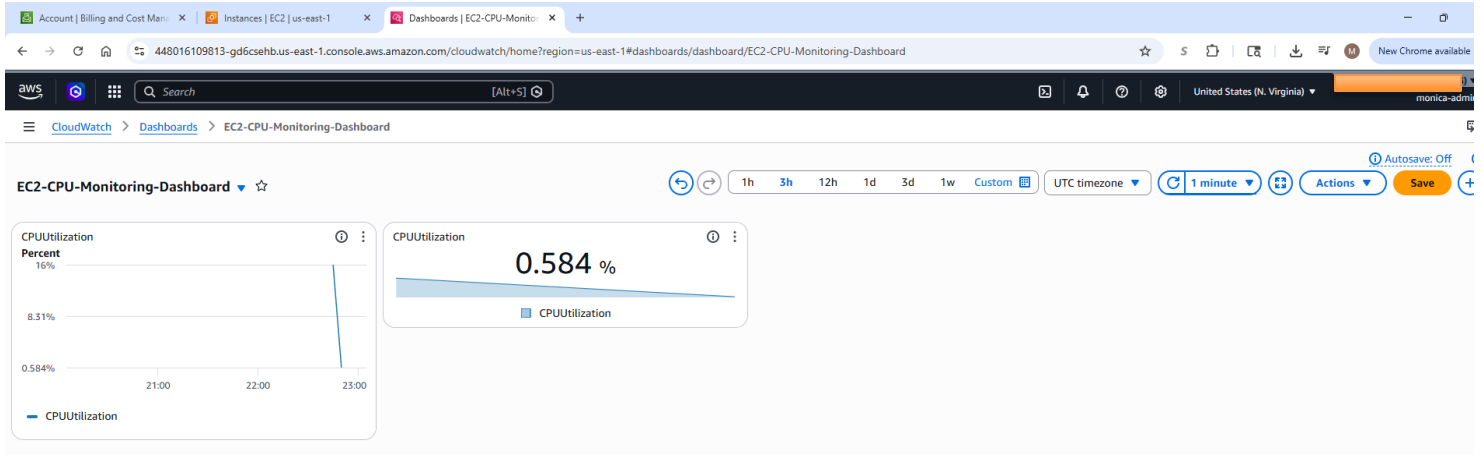


Figure 3 — Simulated CPU utilization spike triggering CloudWatch monitoring thresholds.

A CPU stress test was performed to intentionally increase utilization metrics with **CPUUtilization > 20%** and validate monitoring behavior.

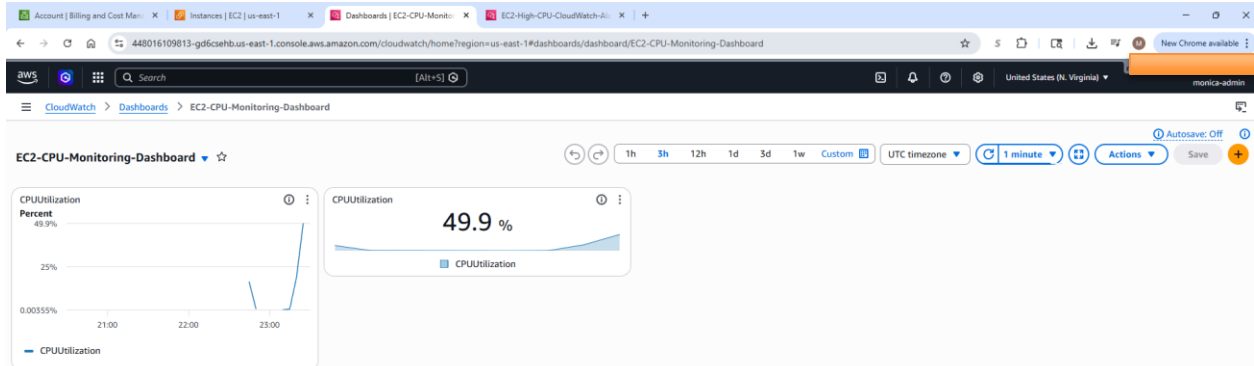
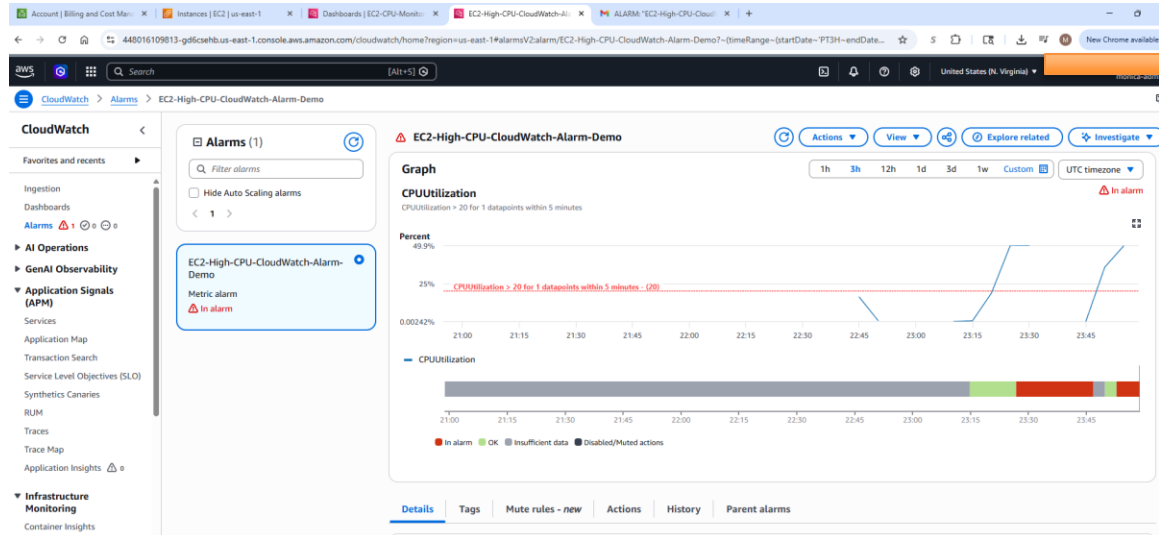


Figure 4 — CloudWatch alarm entering ALARM state after CPU threshold exceeded.

The CloudWatch alarm successfully detected elevated CPU utilization and transitioned into the ALARM state.

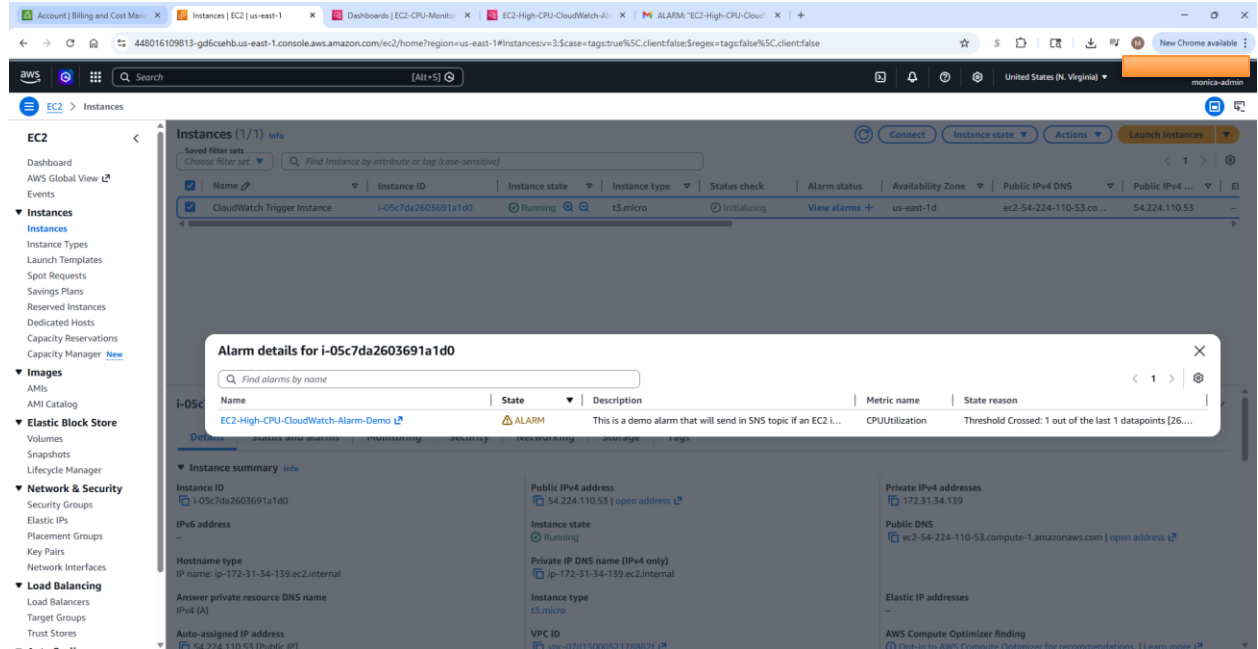
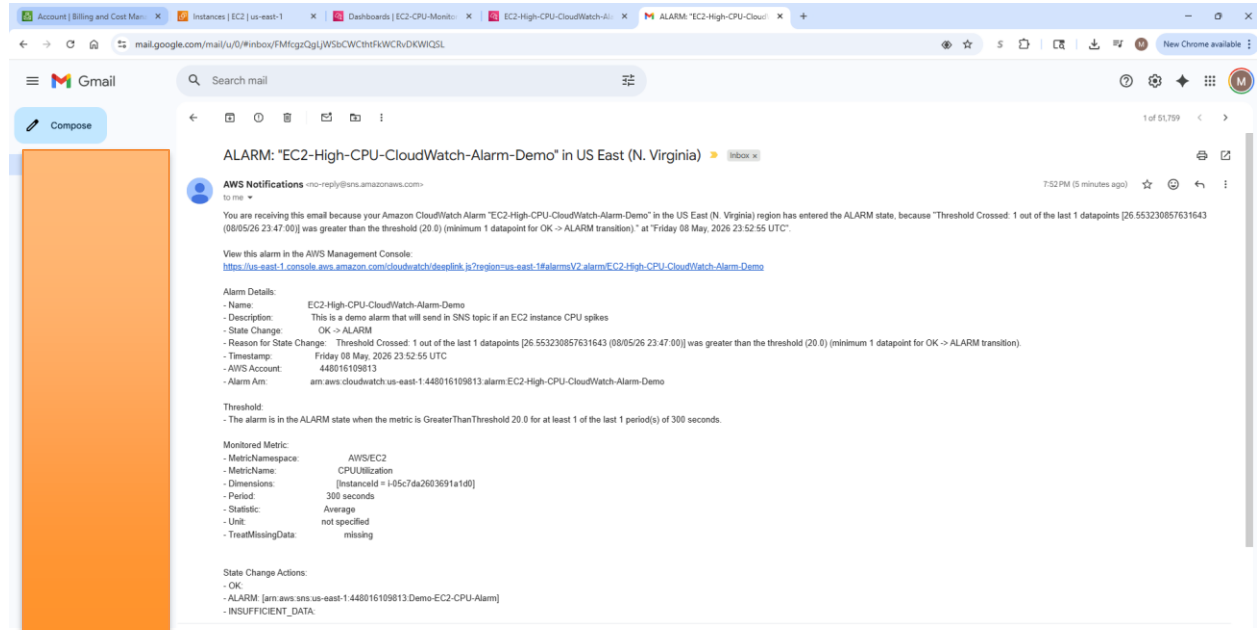


Figure 5 — Amazon SNS email notification triggered from CloudWatch alarm.

Amazon SNS successfully delivered an operational alert email after the EC2 alarm entered the ALARM state.



AWS Services Used

- Amazon EC2
- Amazon CloudWatch
- Amazon SNS

Concepts Demonstrated

- Infrastructure monitoring
- CloudWatch dashboards and metrics
- Operational alerting workflows
- SNS notification delivery
- Incident visibility and response
- Cloud operations monitoring